

iFood

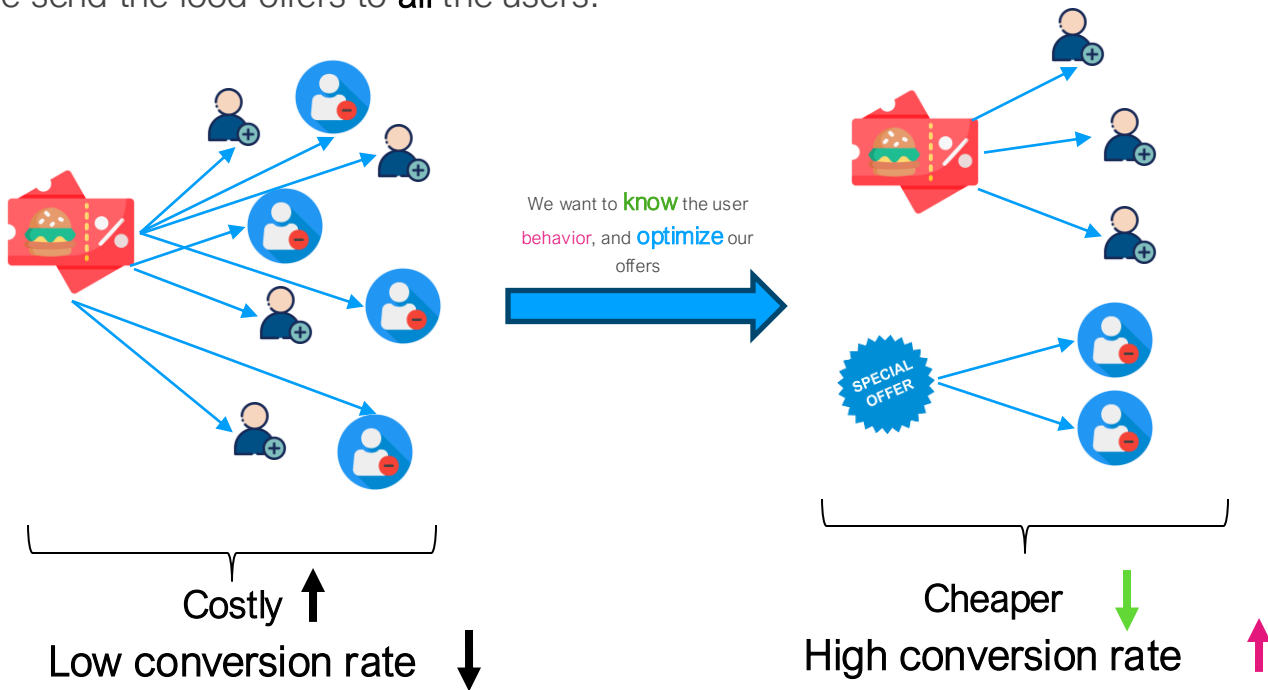


Marketing offer optimization



Problem

Currently, we send the food offers to **all** the users.



Before

50.6k offers → **41.5k** used

EDA Data:

82% conversion rate

12.6k offers → **10.3k** used

Evaluation Data:

81% conversion rate

Now: targeted users

~~12.6k~~ 7.4k offers → 6.8k used

Evaluation Data:

92% conversion rate

Marketing cost:

R\$522.00 saved

Ifood has 50MM users

Marketing campaign

1MM users

414,678 saved marketing messages (SMS or E-mail)

$414,678 \times \text{R\$}0.1 + 414,678 \times \text{R\$}0.004 =$ **R\$43,126.0**

saved

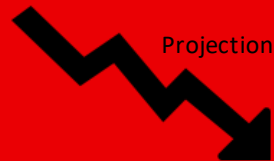
Now: targeted offers

Evaluation Data: **12.6k**

Randomly selected offers



92% conversion rate



67.1%

Exploratory Data Analysis

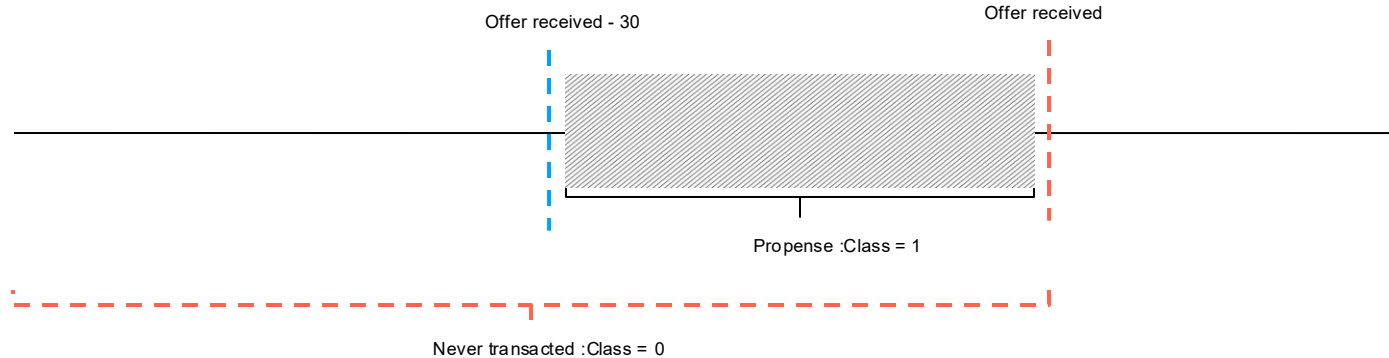


Probably the Data were Generated Sintetically

- Some data are inconsistente or wierd. There are offers that are completed with `time_since_test_start = 0`
- Ages of 120 Years
- Credit card with mean limit value of 60,000

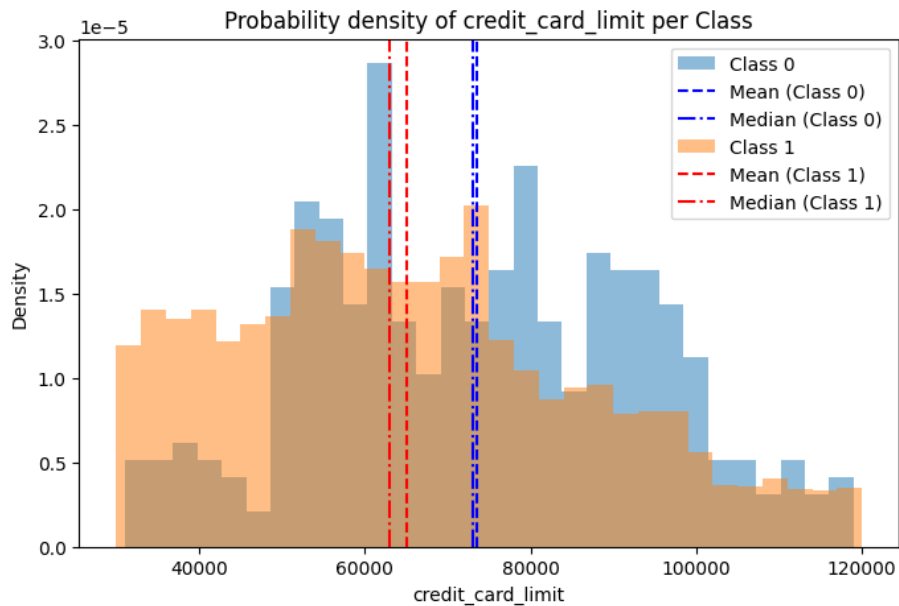
User profile – without offer

“Propense” user definition



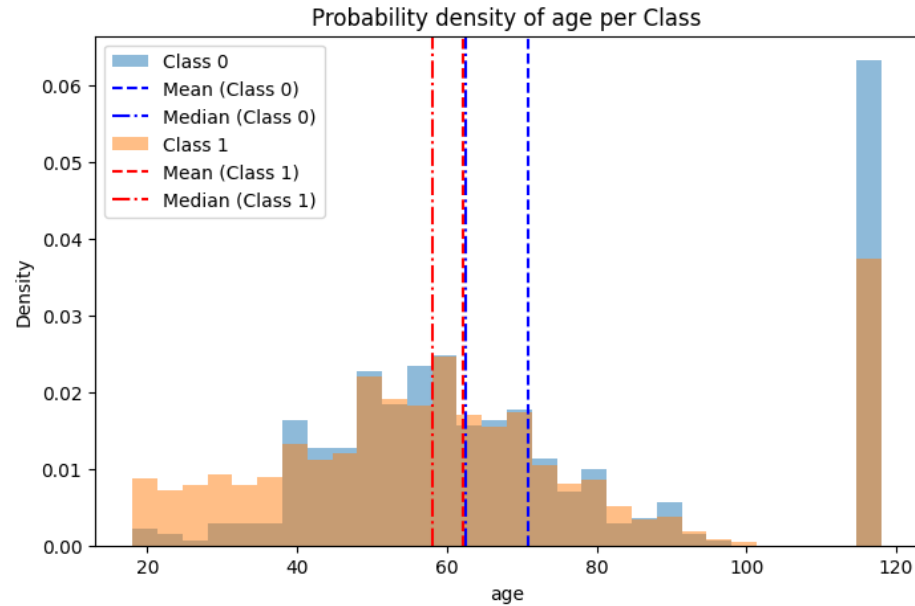
User profile – without offer

Credit Limit and Conversion: mid class uses the application more frequently



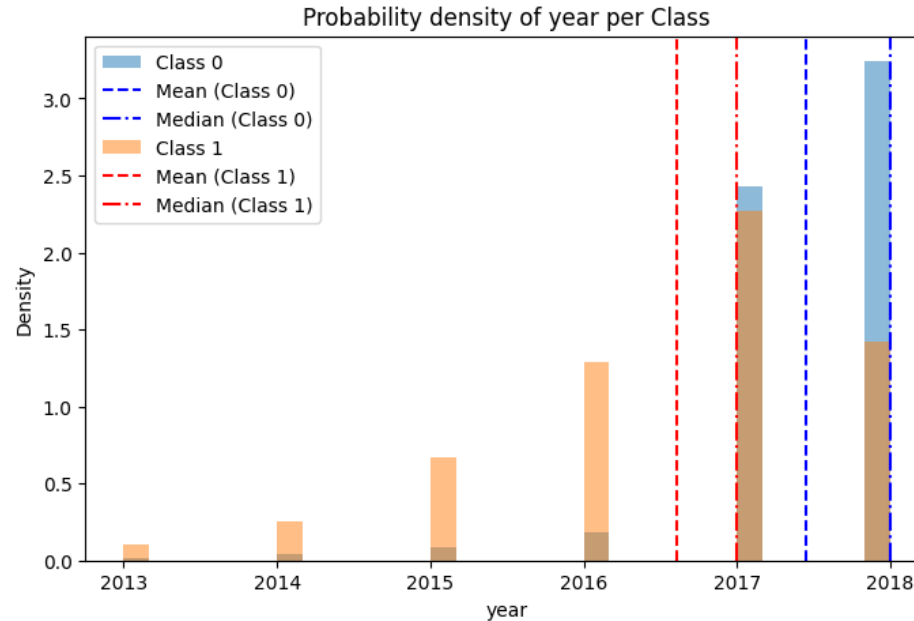
User profile – without offer

Age and Conversion: youngsters transact more



User profile – without offer

Time of using the application: newer users transact less often than older ones

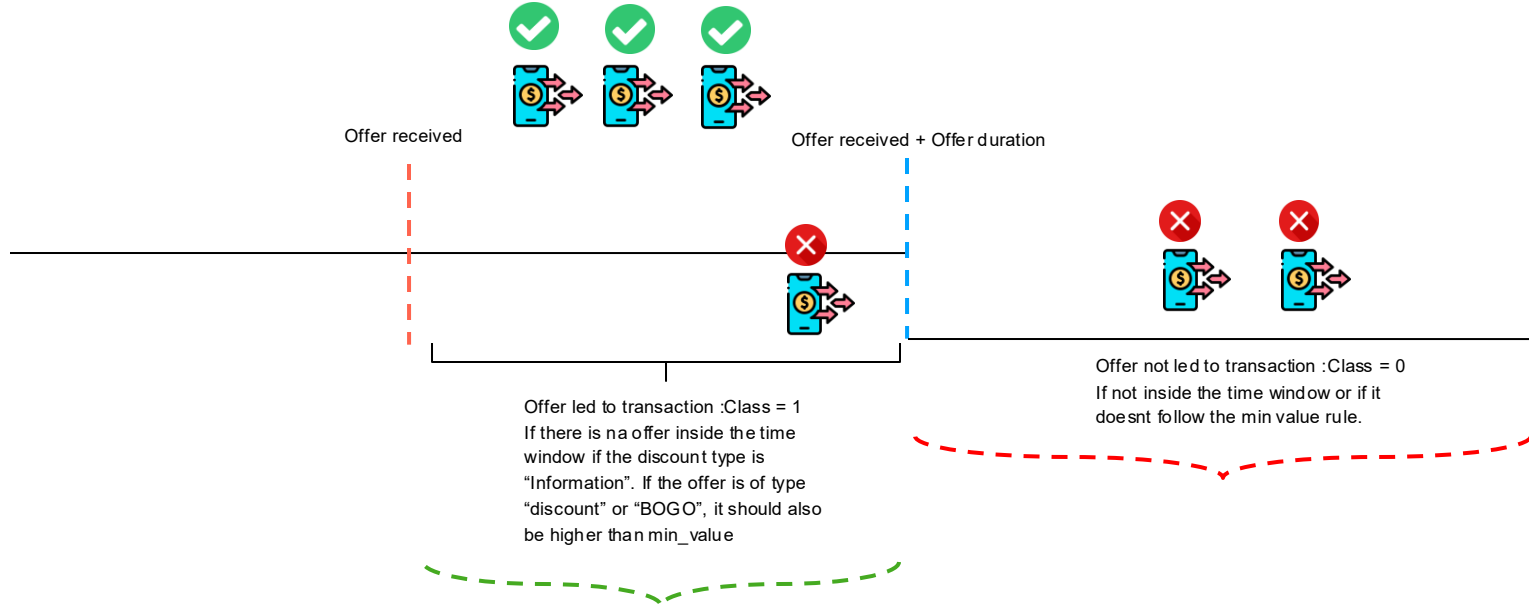


User profile – without offer

Gender: plays no rule

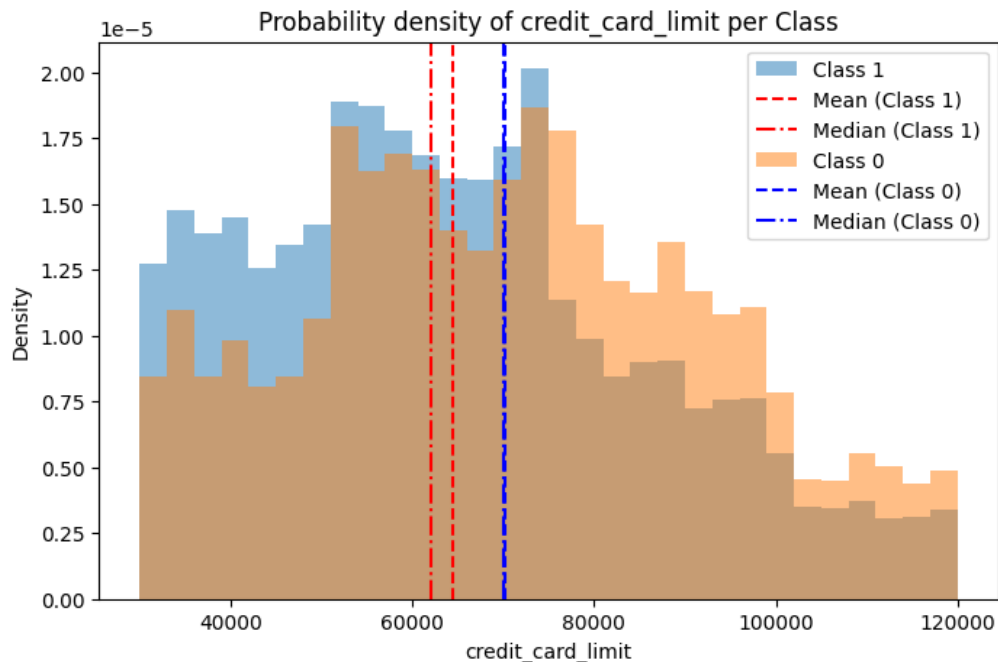
User profile – with offer

“Propense” after receiving an offer



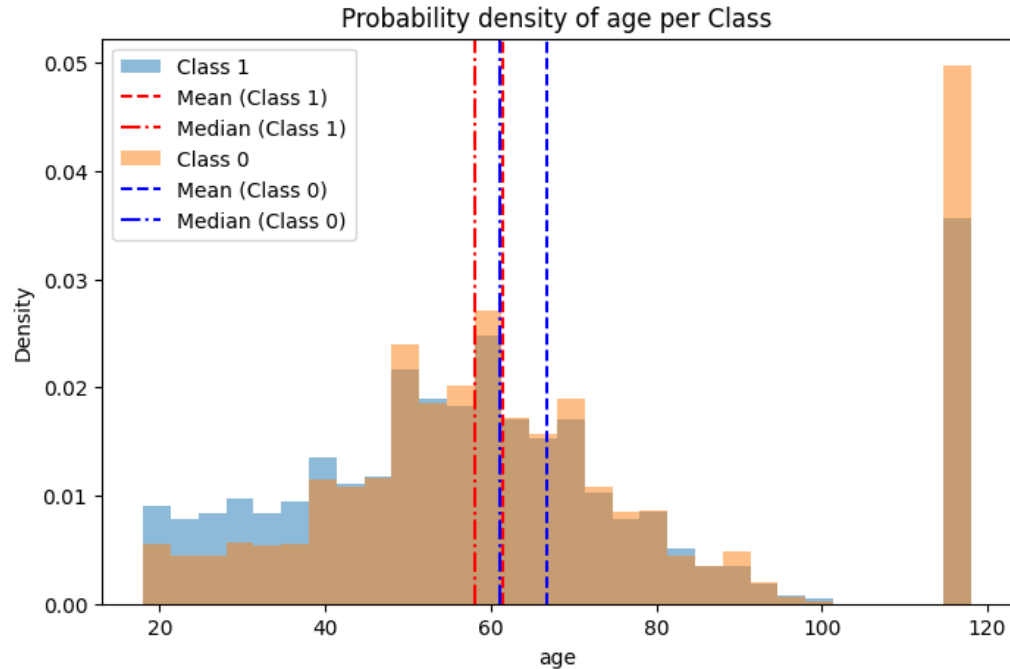
User profile – offer received

Credit Limit and Conversion: mid class uses offers more frequently



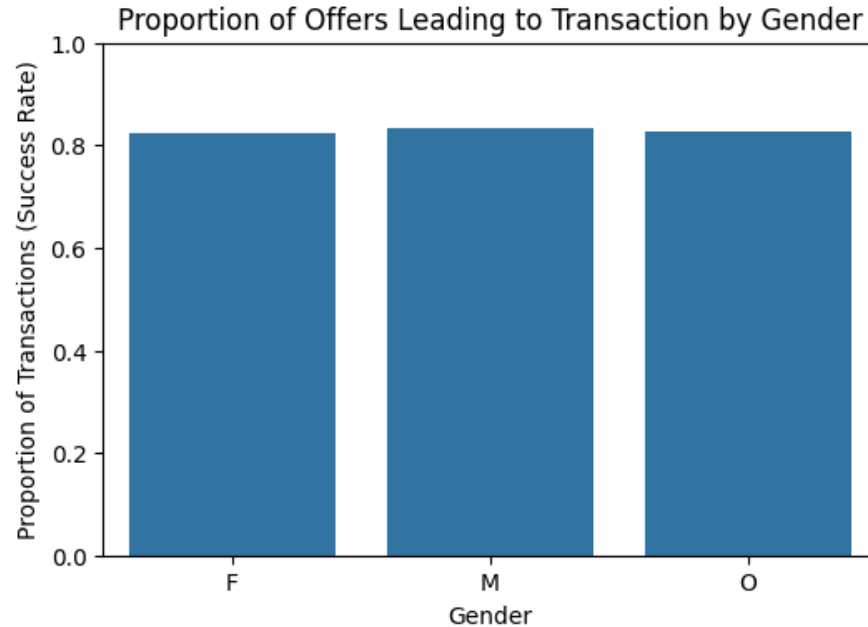
User profile – offer received

Age and Conversion: youngsters accept offers better



User profile – offer received

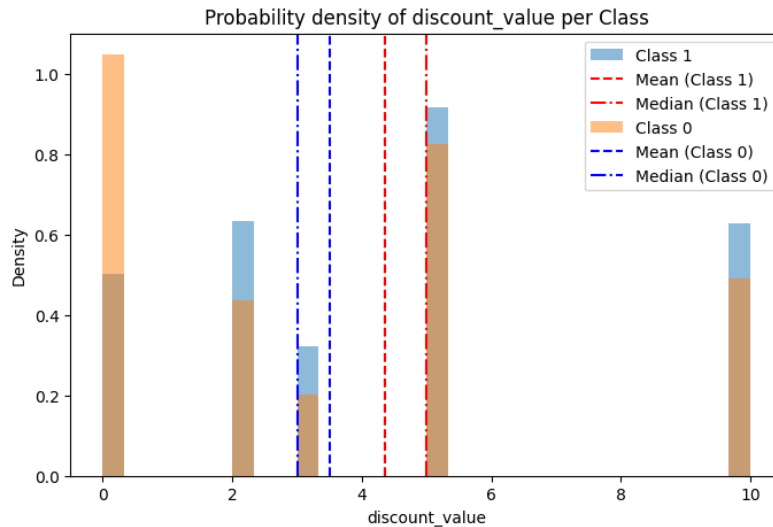
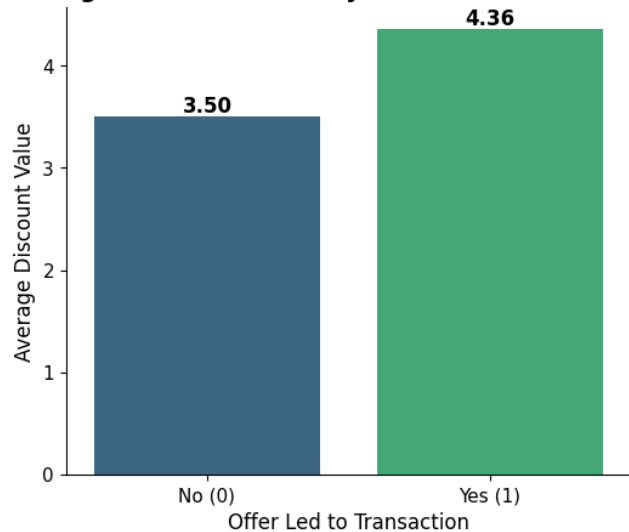
Gender: again no rule at all



Offer characteristics

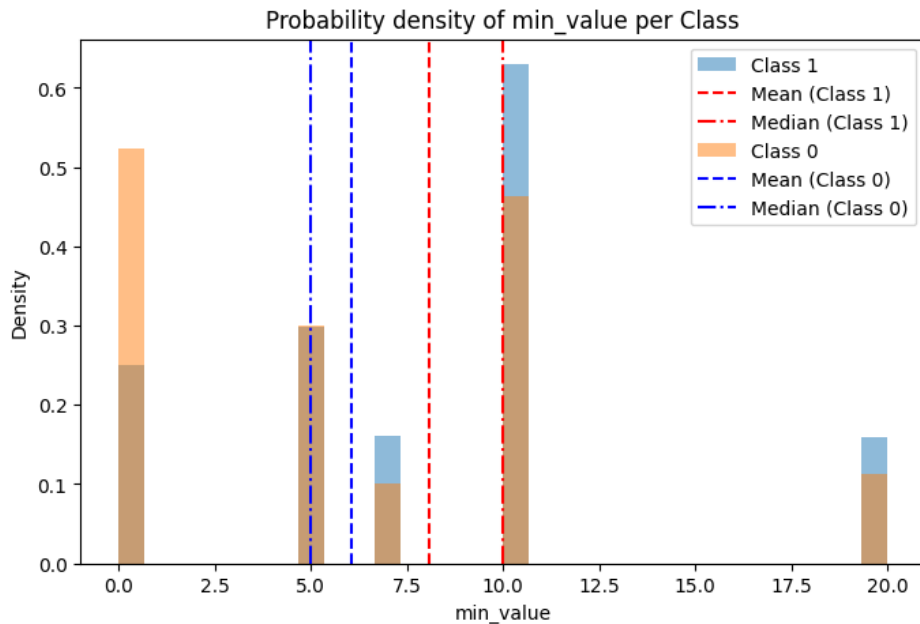
Discount value: offers with higher discount value are more likely led to transaction

Average Discount Value by Offer Led to Transaction



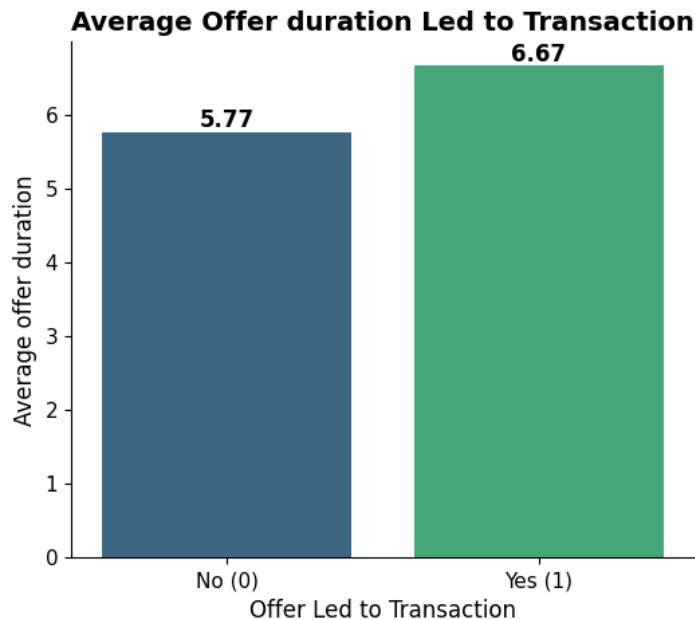
Offer characteristics

Min value: offers with higher min value result more often to a transaction



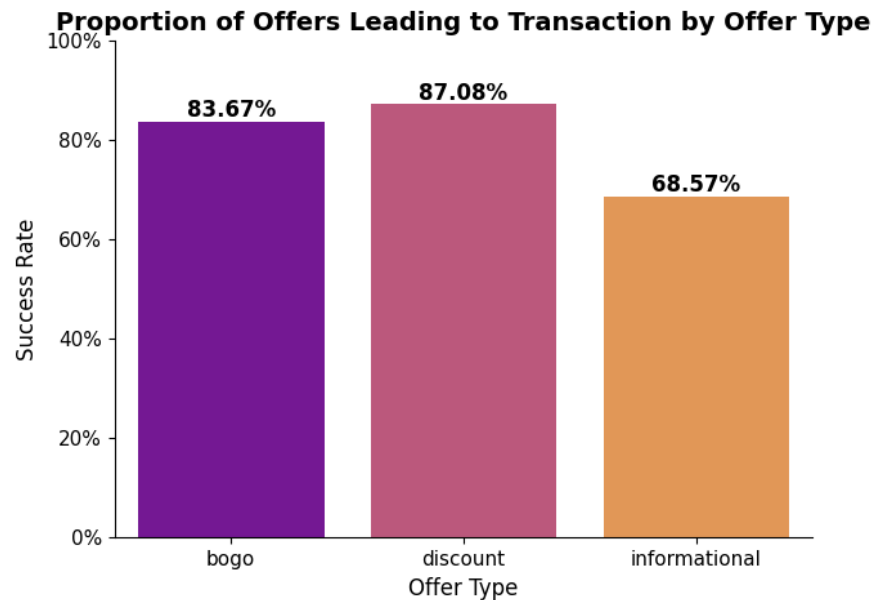
Offer characteristics

Offers duration: longer offers are more likely led to transaction



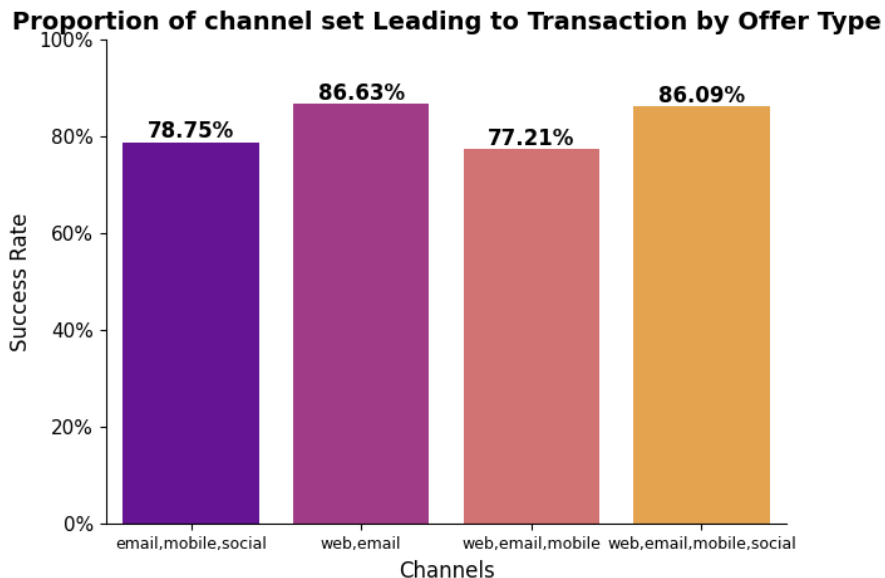
Offer characteristics

Offers type: BOGO and Discount lead to transaction more likely



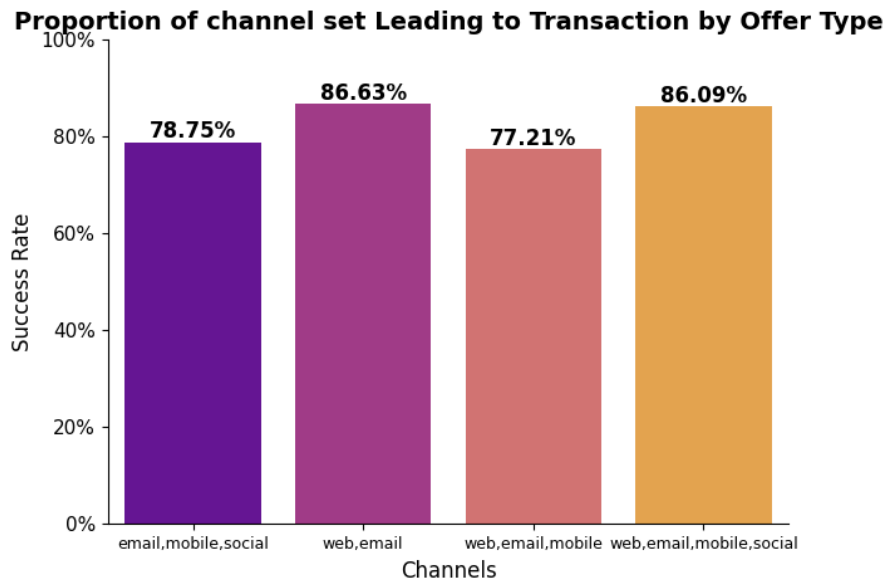
Offer characteristics

Offers channels: web and e-mail results in a higher transaction rate, mobile offers (SMS pushes) are not accepted so well



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Recommendations to Marketing Team



Choosing the user

- **Do not send offer to everyone**
- **Target your user**
- **Define Marketing Goal**
 - Goal: new transaction -> Send offers with high value and long duration for users who never transacted
 - Goal: more transactions -> Send offers to those with a high probability of transaction



Choosing offer type

For each user:

1 - Find the probability of the transaction by changing the offer type

- $P(\text{transaction} \mid \text{offer type} = \text{'BOGO'})$
- $P(\text{transaction} \mid \text{offer type} = \text{'Discount'})$
- $P(\text{transaction} \mid \text{offer type} = \text{'Information'})$

2 - Choose offer type which resulted in a higher transaction probability



Causality

Some users transact anyway

- Use the uplift model. Send the offers for those users who:

Probability of transacting with offer – Probability of transacting without offer $\geq$ some predefined threshold

Obrigado!

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